

Critical Reasoning

Complete Algorithms for CLAT and AILET

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1 Introduction

1.1 What this document is

This manual is a step-by-step guide to the critical reasoning skills tested in CLAT and AILET. It assumes you know nothing. Every technical term is defined before it is used, every skill is turned into a numbered procedure (an **algorithm**, meaning a fixed sequence of steps that you can follow without having to improvise), and most procedures come with a worked example, a list of traps, and a short “exam-speed” version that you will eventually run in your head.

1.2 What this document is not

It is not a guarantee. No procedure is literally foolproof, because language is ambiguous and exam-setters deliberately write tempting wrong options. These procedures make your errors *systematic*: when you get a question wrong, you can point to the step that failed and fix that step. That is the real benefit.

It is also not proven to be the best possible method. The procedures here are standard techniques, adapted from the way logical reasoning is commonly taught for the LSAT, GMAT and CAT-style exams. I am not aware of rigorous evidence showing that these exact procedures beat every alternative. The strongest predictor of improvement is probably the *volume of practice combined with careful review of mistakes*, which is why Algorithm 20 (the review loop) matters more than any single technique.

1.3 The exam in numbers

Exam patterns change, so always confirm against the official notification. The following reflects the CLAT UG 2027 pattern as reported by several coaching and information sources at the time of writing.

Feature	CLAT UG 2027 (reported)
Questions and marks	120 questions, 120 marks
Duration	2 hours (120 minutes), so about one minute per question on average
Mode	Offline, pen and paper, answers marked on an OMR sheet
Marking	+1 for a correct answer, −0.25 for a wrong answer, 0 for an unattempted question

Feature	CLAT UG 2027 (reported)
Format	Multiple-choice questions, all based on reading passages (a passage is often around 450 words)
English Language	About 22–26 questions
Current Affairs including General Knowledge	About 28–32 questions
Legal Reasoning	About 28–32 questions
Logical Reasoning	About 22–26 questions
Quantitative Techniques	About 10–14 questions, at roughly Class 10 level

Legal Reasoning and Current Affairs together carry about half the paper, and Logical Reasoning plus English add roughly another 40 per cent. Most of the algorithms in this manual therefore serve Logical Reasoning, Legal Reasoning and English, while Algorithms 18 and 19 cover Quantitative Techniques and Current Affairs.

For AILET, one comparison source lists 150 questions for the UG paper, but I have not verified its sections or marking scheme. Check the official bulletin at nludelhi.ac.in before relying on any AILET number, and recompute the guessing rule in Algorithm 17 using AILET's actual penalty.

1.4 How to use this manual: the three layers

- (1) **Learn it slowly (layer 1).** For your first 50 or so passages, run each algorithm on paper, writing out every step. This is slow on purpose. You are building the habit.
- (2) **Compress it (layer 2).** Each algorithm has an *exam-speed box* with a three-line version. Once the long version feels obvious, practise only the short version.
- (3) **Run it at exam speed (layer 3).** With one minute per question, you cannot write out steps. The compressed version must run silently in your head, and you fall back on the long version only when a question confuses you.

1.5 Honest limitations and known corrections

Earlier drafts of these procedures contained simplifications. They are fixed in this document, and you should know about them because other books and coaching notes make the same simplifications.

- (1) **Speed.** The written algorithms are training tools. Do not try to write them out in the exam.

- (2) **The deletion test for premises is only a rough check.** If two premises independently support a conclusion, deleting one leaves the conclusion supported, so the test wrongly calls it a non-premise. Use the “because” test first (Algorithm 2).
- (3) **The negation test works only for *necessary-assumption* questions.** It does not work for strengthen, weaken or sufficient-assumption questions (Algorithms 5 and 6).
- (4) **“All P are Q , therefore some Q are P ” assumes that P is not empty.** Exams normally treat it as valid, but you should know the hidden assumption (Algorithm 11).
- (5) **Heuristics are not laws.** Rules such as “choose the milder tone”, “choose the narrower inference” and “the right assumption is usually modest” are tiebreakers, to be used only after the real checks leave two options standing.
- (6) **The fallacy list over-simplifies.** Appealing to an expert is legitimate evidence (though not proof). Questioning a person’s credibility is sometimes relevant. And a flawed argument does not mean its conclusion is false (Algorithm 7).
- (7) **“Counting arguments” depends on the exact wording of the question.** The convention in Algorithm 3 is sensible, but I am not sure how often this is tested, so always follow the stem.

1.6 Conventions used in this manual

- **Blue boxes** give definitions.
- **Green boxes** give worked examples.
- **Red boxes** list traps and common mistakes.
- **Orange boxes** give the exam-speed version of an algorithm.
- **Grey boxes** give notes and caveats.
- P, Q, R stand for statements, and A, B, C stand for groups of things. The symbol $\neg P$ means “not P ”, and $P \rightarrow Q$ means “if P then Q ”.

2 Foundation Vocabulary

Read this section first. Every algorithm that follows depends on these terms.

2.1 The basic building blocks

Term	Definition	How to recognise it
Passage	The block of text you are given to read.	The paragraph(s) above the questions.
Statement	A complete sentence that says something that could be true or false.	“The river flooded.” Questions and commands are not statements.
Fact	A statement that can in principle be checked and shown true or false.	Dates, numbers, events: “The Constitution came into force in 1950.”
Opinion or claim	A judgment or belief on which reasonable people may disagree.	Words such as <i>should</i> , <i>ought</i> , <i>best</i> , <i>worst</i> , <i>unfair</i> , <i>harmful</i> .
Argument	A set of statements in which some (the premises) are offered to convince you to accept another (the conclusion).	A passage that tries to persuade rather than merely inform.
Conclusion	The statement the author wants you to accept.	Often an opinion, recommendation, prediction or evaluation.
Premise	A statement given as a reason to accept the conclusion.	It answers “why should I believe this?”
Evidence	A fact, statistic, example or expert statement used as a premise.	Numbers, studies, named cases.
Assumption	An unstated premise that must be true for the argument to work.	It never appears in the text; you work it out.
Inference	A statement that follows from the passage by logic.	Not written in the passage, but forced or strongly supported by it.
Tone	The author’s attitude to the subject, shown by word choice.	Evaluative words and phrasing.
Counter-argument	A view the author mentions in order to oppose it.	“Critics argue...”, followed by a rebuttal.
Concession	A point the author admits is valid before making the real point.	“Admittedly...”, “It is true that...”, followed by <i>but</i> .

2.2 Types of statements

Different kinds of statements behave differently in an argument, and many wrong options come from mixing them up.

- (1) **Descriptive (factual) statement:** says what is the case. “Air pollution in the city exceeds safe limits.”
- (2) **Evaluative statement:** says whether something is good or bad. “The scheme has been a failure.”
- (3) **Prescriptive statement (recommendation):** says what should be done. “The scheme should be scrapped.”
- (4) **Predictive statement:** says what will happen. “Fares will rise next year.”
- (5) **Definitional statement:** says how a word is being used. “By *poverty* we mean income below a fixed line.”

A passage that only contains descriptive statements cannot, by itself, support a prescriptive conclusion. Something must bridge the gap, and that missing bridge is usually an **assumption** (Algorithm 5).

2.3 Indicator words

Indicator words are signposts that tell you what role a sentence is playing. Memorise them in groups.

Group	Words	What it tells you
Conclusion indicators	<i>therefore, thus, hence, so, consequently, it follows that, accordingly, as a result, clearly, in conclusion, which shows that</i>	The next claim is probably the conclusion.
Premise indicators	<i>because, since, for, given that, as, owing to, due to, in view of, the reason is, as shown by, studies show</i>	The next claim is probably a reason.
Contrast indicators	<i>but, however, yet, although, though, whereas, nevertheless, on the other hand, despite, in contrast, still</i>	The direction of thought changes. The author's real view often follows.
Addition indicators	<i>moreover, furthermore, in addition, also, besides</i>	More support is coming.
Example indicators	<i>for example, for instance, such as, consider</i>	What follows illustrates; it is not the main point.
Counter-view indicators	<i>critics claim, some argue, it is said, it is often believed, according to X, proponents say</i>	The author is reporting someone else's view, not necessarily their own.

Indicator words are clues, not proof

Since can mean *because* (a premise indicator) or *from the time that* (a time word). As can mean *because*, *while* or *like*. Always test the sentence rather than trusting the word.

2.4 Quantifiers and modals

Many wrong options are wrong only because they change the *strength* of a word.

Quantifiers say *how many*.

Word	Meaning
all, every, each	100 per cent
most, majority	more than 50 per cent
many, several	a notable number, not specified
some, a few, at least one	one or more, possibly all
none, no	0 per cent

Modals say *how certain*.

Word	Strength
must, will, certainly	very strong
should, ought	a recommendation
likely, probably	more than 50 per cent chance
can, may, might, could	possibility only

The golden strength rule

A valid inference may be *equal to or weaker than* what the passage says, never stronger. “Some” cannot become “all”. “May” cannot become “will”. “Is linked to” cannot become “causes”.

2.5 Types of reasoning

- **Deductive reasoning:** if the premises are true, the conclusion *must* be true. “All judges are lawyers. Meera is a judge. So Meera is a lawyer.”
- **Inductive reasoning:** the premises make the conclusion *probable*. “Every swan I have seen is white, so swans are probably white.”
- **Causal reasoning:** claims that one thing causes another.

- **Analogical reasoning:** argues that because A and B are alike, what is true of A is true of B .

2.6 How premises support a conclusion

Three patterns appear again and again. They matter for Algorithms 2 and 3.

- **Convergent support:** several premises each support the conclusion independently. If you removed one, the others would still help.
- **Linked support:** premises that work only together. Neither supports the conclusion alone.
- **Chain (sub-argument):** premise $\rightarrow$ intermediate conclusion $\rightarrow$ main conclusion.

2.7 The two ground rules of exam reasoning

Rule 1: Who is speaking?

Always identify the speaker. A statement may belong to the author, to a person quoted, to an opponent whom the author is about to rebut, or to a historical figure being described. Many wrong options attribute a quoted person's view to the author.

Rule 2: The passage is the whole world (in the exam)

In Logical Reasoning, Legal Reasoning and most English questions, treat the passage as the entire evidence. Do not import outside knowledge, even if it is true. (In real life the opposite holds; see Section 24.)

3 The Master Pre-Step

Do this for every passage before applying any algorithm. Its purpose is to read the passage as a *structure* rather than a wall of words.

3.1 What you are looking for

- **Topic:** what the passage is about (a noun phrase, such as “judicial review of legislation”).
- **Main point:** what the author *says* about the topic (a full sentence).
- **Structure:** how the paragraphs relate, for example problem then solution, claim then objection then reply, or history then analysis.

3.2 Procedure

- (1) **Read the question stems first, not the options** (about 20 seconds). This tells you whether to hunt for tone, an assumption, an inference or something else.
- (2) **Read the passage once at normal speed.** While reading, circle contrast indicators and conclusion indicators.
- (3) **After each paragraph, write a five-word gist** in the margin (for example, “Cause of river pollution”).
- (4) **After the last paragraph, write the main point** in one sentence: “The author argues that ...”
- (5) **Mark who is speaking.** Is each claim the author’s, a quoted person’s, or an opponent’s?
- (6) **Mark the strength words** (quantifiers and modals) in sentences that might be tested.

Exam-speed version

Stems first. One read, circling *but/therefore*. Five-word gist per paragraph. One-line main point. Then answer.

Why stems first?

Reading the questions first is a common recommendation, but it has a cost: if you read stems carefully and then hold six questions in your head, you may read the passage anxiously and miss its overall flow. Try both

approaches on practice papers and keep whichever gives you higher accuracy. A middle path is to skim only the *question types* (tone, inference, main idea) rather than every detail.

4 Algorithm 1: Identifying Tone

4.1 Definitions and parameters

Tone

Tone is the author's attitude toward the subject, as revealed by word choice. It is *not* the topic (a sad topic can be described in a detached tone) and *not* the mood (the feeling the passage produces in the reader).

Every tone has four parameters. Determine each of them before looking at the options.

- (1) **Target:** what the attitude is directed at (a policy, a person, a trend, an idea). A passage may have several targets with different attitudes, so find the one the question asks about.
- (2) **Polarity:** is the author's attitude positive, negative, neutral or mixed?
- (3) **Intensity:** is it mild, moderate or strong?
- (4) **Function:** what is the author *doing*: informing, criticising, praising, warning, urging, mourning, mocking?

4.2 How to detect polarity and intensity: the ten clues

- (1) **Diction (word choice).** Evaluative words carry the attitude. Positive: *remarkable, commendable, visionary, thriving, landmark*. Negative: *flawed, dismal, reckless, hollow, deplorable*. Neutral: factual verbs like *states, reports, notes, describes*.
- (2) **Connotation.** Two words can mean nearly the same thing but feel different. *Firm* is positive, *stubborn* is negative, *obstinate* is more strongly negative.
- (3) **Hedging words** (*perhaps, arguably, somewhat, seems*) signal a measured or tentative tone.
- (4) **Intensifiers** (*utterly, wholly, absolutely, grossly*) signal strong feeling.
- (5) **Scare quotes and "so-called".** Putting a word in quotation marks, or writing *so-called*, signals doubt or mockery.
- (6) **Rhetorical questions.** A question asked without expecting an answer ("Is this really progress?") usually signals criticism.

- (7) **Irony.** The author says the opposite of what they mean. Test: does the literal statement make sense given the facts the passage supplies? If it clearly does not, the author is being ironic.
- (8) **Punctuation.** Exclamation marks signal strong feeling. Dashes and parentheses often add asides or sarcasm.
- (9) **Sentence rhythm.** Short, blunt sentences feel urgent or angry. Long, balanced sentences feel analytical.
- (10) **Absence of loaded words.** If nearly every word is factual, the tone is neutral or objective. This is the most common correct answer in real exams.

4.3 The tone bank, with meanings

Learn each word's *meaning*, not just the word.

Positive tones

Tone	Meaning
Appreciative	Recognises and values something
Laudatory	Praises highly
Admiring	Respects and looks up to
Optimistic	Expects good outcomes
Hopeful	Wishes and sees a possibility of a good outcome (milder than optimistic)
Enthusiastic	Eager, energetic approval
Approving	Considers something acceptable or good
Celebratory	Rejoices over an achievement
Reverential	Shows deep respect, close to awe

Neutral tones

Tone	Meaning
Objective	Based on facts, without personal feeling
Analytical	Breaks a topic into parts to examine it
Informative, expository	Explains or gives information
Descriptive	Paints a picture of what something is like
Dispassionate, detached	Emotionally uninvolved
Impartial	Favours no side
Measured	Careful, balanced, restrained
Reflective	Thoughtfully considers or looks back

Tone	Meaning
Inquisitive	Curious, asking questions

Mildly negative tones

Tone	Meaning
Critical	Points out faults
Sceptical	Doubts claims and wants proof
Cautionary	Warns gently of possible harm
Concerned	Worried
Apprehensive	Anxious about the future
Disapproving	Considers something wrong
Dismissive	Treats something as unworthy of consideration
Ambivalent	Holds mixed or conflicting feelings
Pessimistic	Expects bad outcomes
Regretful	Feels sorry about the past
Wistful	Gently sad longing for something gone or out of reach
Resigned	Accepts something unpleasant without resisting

Strongly negative tones

Tone	Meaning
Scathing	Harshly, severely critical in manner
Sarcastic	Uses mocking, often ironic remarks meant to wound
Satirical	Uses humour, irony or exaggeration to expose foolishness in society or institutions
Sardonic	Grimly or bitterly mocking; dry, cynical amusement
Acerbic	Sharp and bitter in speech
Indignant	Angry at something seen as unjust
Condemnatory	Declares something strongly wrong
Derisive	Treats with contempt and ridicule
Cynical	Believes people act only from selfish motives
Polemical	Argues aggressively against a position

Directive (action-oriented) tones

Tone	Meaning
Exhortatory	Strongly urges people to act
Admonitory	Warns or gently scolds
Urgent	Says something must be done now
Prescriptive	Lays down what should be done
Advocating	Argues in support of a cause
Alarmist	Exaggerates danger to frighten

Tone	Meaning
Didactic	Teaches, often in a moralising manner

Other tones

Tone	Meaning
Nostalgic	Fond, sentimental longing for the past
Elegiac	Mournful, like a lament for something lost
Apologetic	Expresses regret for a fault
Defensive	Protects itself against criticism
Conciliatory	Tries to make peace
Whimsical	Playfully odd
Ironic	Meaning the opposite of the literal words

4.4 Pairs that students confuse

Pair	The difference
Critical vs. cautionary	Critical points out faults that exist now. Cautionary warns about harm that may come.
Objective vs. analytical	Objective means no personal feeling. Analytical means breaking something into parts. An analytical passage is usually objective, but an objective passage need not analyse.
Ironic, sarcastic, satirical, sardonic	Irony is a technique (saying the opposite of what is meant). Sarcasm is irony aimed to wound. Satire is sustained mockery of society or institutions. Sardonic is dry, bitter, grimly amused.
Ambivalent vs. impartial	Ambivalent means the author is torn and feels both ways. Impartial means the author takes no side and shows no feeling.
Nostalgic, wistful, elegiac	Nostalgic is fond longing for the past. Wistful is lighter, tinged with regret. Elegiac is mournful, a lament.
Sceptical vs. cynical	Sceptical doubts claims until given evidence. Cynical assumes bad motives.
Optimistic vs. hopeful	Optimistic expects success. Hopeful merely wishes and sees a possibility.
Indignant vs. scathing	Indignant names an emotion (anger at injustice). Scathing names a manner (severely critical writing).
Didactic vs. prescriptive	Didactic describes a teaching or moralising manner. Prescriptive describes laying down what should be done.
Apprehensive vs. alarmist	Apprehensive is genuinely worried. Alarmist exaggerates the danger to frighten.

4.5 The algorithm

- (1) **Identify the target.** Find what the question asks about (the author's attitude to "the policy", or to "judicial activism"). Ignore other targets.
- (2) **Confirm the speaker.** Is it the author or a quoted person? The question almost always asks about the author.
- (3) **Underline every evaluative word** aimed at the target: adjectives, adverbs, loaded verbs, scare quotes.
- (4) **Set polarity.** Count positive against negative words. If mixed, look at what comes *after* the last contrast word (*but, however*), because that usually carries the author's real view.
- (5) **Set intensity.** Are the words mild (*problematic, unfortunate*) or strong (*disastrous, appalling*)? Few or no loaded words means mild or neutral.
- (6) **Set function.** Is the author informing, criticising, praising, warning, urging or mocking? This separates, for example, *critical* from *cautionary*.
- (7) **Pick from the bank** the tone that matches polarity, intensity and function together.
- (8) **Eliminate:**
 - (a) any option with the wrong polarity;
 - (b) any option stronger than the text supports ("outraged" for a mildly worded text);
 - (c) "sarcastic", "satirical" or "ironic" unless the text clearly says the opposite of what it means;
 - (d) a tone that belongs to a quoted person rather than the author;
 - (e) tones for which you cannot underline a single justifying word.
- (9) **Final test.** Can you point to two words in the passage that justify your chosen tone? If not, reconsider.

A tiebreaker, not a law

If two options both fit after step 9, prefer the *milder* one that matches the text, unless you can point to specific evidence (a scare quote, an ironic contrast) for the more specific, stronger tone. Examiners rarely offer two options that are equally defensible, so if you are stuck between two, you have probably misjudged polarity or intensity.

4.6 Worked examples

Example 1: irony

“The government’s much-touted scheme has, after three years and considerable expense, reached barely four per cent of its intended beneficiaries. One wonders what was being measured when officials declared it a triumph.”

Options: (a) Appreciative (b) Dispassionate (c) Sardonic (d) Indignant.

Steps. Target: the government scheme. Speaker: the author. Evaluative words: *much-touted* (mocking), *barely*, *one wonders*, *declared it a triumph* set against a four per cent result. Polarity: negative. Intensity: moderate. Function: mocking the gap between claim and result.

Elimination. (a) wrong polarity. (b) wrong, since there are loaded words. (d) indignant would need signs of anger at injustice (exclamation marks, words such as *outrage*); the dry phrase *one wonders* is wry, not angry.

Answer: (c) Sardonic. The author never says *failure*; the contrast between *triumph* and *four per cent* does the mocking. If *critical* had been offered instead of *indignant*, you would choose *sardonic* only because of the two pieces of evidence (the scare-like *much-touted* and the ironic *declared it a triumph*); without them *critical* would be the safe choice.

Example 2: a concession followed by a contrast

“Admittedly, the new rules reduce paperwork, and officials deserve credit for that. But the rules also remove the one safeguard that let citizens appeal against arbitrary decisions, and no amount of efficiency justifies that.”

Options: (a) Appreciative (b) Critical (c) Ambivalent (d) Indignant.

Steps. The first sentence is positive, but it is a concession (*Admittedly*). The weight lies after *But*. The closing clause, *no amount of efficiency justifies that*, is firm and negative, but it is expressed calmly.

Elimination. (a) wrong, because the praise is only a concession. (c) ambivalent would mean the author is genuinely torn; here the author is not torn, since the conclusion is firmly negative. (d) indignant needs signs of anger, and the language is controlled.

Answer: (b) Critical.

Common mistakes in tone questions

- Choosing the tone of the *topic* (a tragic subject does not mean a tragic tone).

- Choosing the tone of a quoted character instead of the author.
- Choosing an extreme word (*outraged*, *hostile*) when the text is measured.
- Ignoring the last two sentences, where tone often shifts.
- Choosing *sarcastic* because the passage is critical. Sarcasm requires irony.

Exam-speed version

(1) Target and speaker. (2) Polarity and intensity from the loaded words, with the last sentence weighted most. (3) Function. Then choose the option you can back with two words from the text.

5 Algorithm 2: Identifying Premises

5.1 Definitions and parameters

Premise and conclusion

A **premise** is a statement offered as a reason to accept another statement. The **conclusion** is the statement the premises are meant to prove. An **intermediate conclusion** is a statement that is itself supported by premises and also supports the main conclusion, so it plays both roles.

What is *not* a premise.

- **Background:** context that supports nothing.
- **Counter-views:** opposing claims that the author will reject.
- **Pure illustrations:** examples that merely make an idea vivid.
- **The conclusion itself**, or a restatement of it.

Types of premises.

- **Factual premise:** a verifiable fact or statistic.
- **Value premise:** a judgment about what matters (“Public health matters more than convenience”).
- **Definitional premise:** how a term is being used.
- **Authority premise:** an expert says so.
- **Example premise:** a specific case offered as support.

The parameters you must determine are: (a) the main conclusion, (b) the statements that directly support it, and (c) the statements that support those.

5.2 The algorithm

- (1) **Split the passage into separate statements** and number them.
- (2) **Label each statement** as Opinion or Recommendation (O), Fact (F) or Background (B).
- (3) **Find the main conclusion.** Look for conclusion indicators (*therefore, thus, hence*). If there are none, ask: “What is the author trying to make me believe or do?” It is usually a recommendation, prediction or evaluation, and it is often the first or last sentence.

- (4) **Ask “Why does the author say this?”** Each answer is a candidate premise.
- (5) **Check premise indicators** (*because, since, given that, studies show*). The statement after them is likely a premise.
- (6) **Run the “because” test (primary test).** Say: “[Conclusion] because [candidate].” If it reads naturally, the candidate is a premise. If the reverse reads better, you have them swapped.
- (7) **Handle intermediate conclusions.** If the candidate itself needs a reason, it is a sub-conclusion, and its reasons are the premises of a sub-argument.
- (8) **Remove non-premises:** background, counter-views, pure illustrations and restatements.
- (9) **Optional rough check (the deletion test).** Imagine removing the candidate. If the conclusion is now less supported, it was probably a premise. *Caution:* this test fails when the author gives two independent reasons for the same conclusion, because removing one leaves the other. Treat a failed deletion test as a prompt to re-check, not as proof.

In legal reasoning questions

The stated *principle* and the stated *facts* are the premises. Do not add outside law as a premise unless the question tells you to use real law. See Algorithm 13.

5.3 Worked examples

Example 1: a simple argument

“Cities should ban private cars from their centres. Air quality in such areas regularly exceeds safe limits. Studies show that pedestrianised zones raise local shop revenue. Some argue that bans inconvenience commuters, but the evidence on air quality is decisive.”

Conclusion: cities should ban private cars from their centres (a recommendation). Premise 1: air quality exceeds safe limits (fact). Premise 2: pedestrianised zones raise revenue (fact from studies). *Not* a premise: *bans inconvenience commuters*, which is a counter-view reported only to be set aside.

Example 2: a chain with an intermediate conclusion

“The river is polluted, so fish stocks have fallen, and local fishers are therefore losing income. The government should compensate them.”

Main conclusion: the government should compensate the fishers. Intermediate conclusion: fishers are losing income. Its support: fish stocks have fallen. That in turn is supported by: the river is polluted. Note what is *missing*: nothing says the government is responsible for compensating people harmed by pollution. That unstated value premise is an **assumption**, which is the subject of Algorithm 5.

Example 3: why the deletion test can mislead

“The scheme should be scrapped. It has missed every target for three years, and an independent audit found that half its funds were never spent.”

Delete the first reason and the audit still supports the conclusion. Delete the audit and the missed targets still support it. A naive deletion test would call *both* non-premises, which is plainly wrong. The “because” test gives the right result: “The scheme should be scrapped *because* it has missed every target” and “... *because* an audit found half its funds unspent” both read naturally.

Common mistakes in premise questions

- Treating the first sentence as the conclusion because it comes first. Position is a hint, not a rule.
- Treating a counter-view as a premise.
- Treating an example as the premise rather than as an illustration of one.
- Missing that a conclusion can also be the premise of a larger argument.

Exam-speed version

Find the claim the author most wants accepted. Ask “because?” Anything that answers it is a premise; anything that is context, a rejected view or a pure illustration is not.

6 Algorithm 3: Counting the Arguments

6.1 Definitions and parameters

First decide what the question is counting. “How many arguments does the author make?” normally means the number of *distinct conclusions being defended*. It does not mean the number of reasons or the number of sentences. I am not sure how often this exact question appears in CLAT and AILET, so always follow the exact wording of the stem (for example, “arguments presented” may include arguments the author reports and rejects).

Terms used when counting

Independent argument: a conclusion with its own support that does not serve as support for another conclusion.

Chain (sub-argument): premise $\rightarrow$ intermediate conclusion $\rightarrow$ main conclusion. This counts as *one* argument.

Convergent support: several premises each independently supporting one conclusion. This is *one* argument with several reasons.

Linked support: premises that work only together. This is *one* argument.

Opposing view: a view the author reports in order to refute. It is not the author’s argument unless the question asks about “arguments discussed in the passage”.

6.2 The algorithm

- (1) **Read the question carefully.** Does it ask for arguments “made by the author”, “presented”, or “reasons given”? This changes what you count.
- (2) **Number every sentence** and label each as a claim (C), evidence (E) or background (B).
- (3) **List every candidate conclusion.** These are the claims that ask you to believe or do something.
- (4) **Draw the support map.** Draw an arrow from each piece of evidence to the claim it supports, and from each claim that supports another to that other claim.
- (5) **Merge chains.** If claim *A* supports claim *B*, and *B* supports claim *C*, then only *C* is a final conclusion. That is one argument.
- (6) **Remove counter-views** (unless the question includes them).

- (7) **Count the final conclusions** that do not support any other claim. That is your number.
- (8) **Double-check.** If one option says “two” and another says “one”, go back to step 5 and ask whether one of your “conclusions” is really only a stepping stone.

A useful signal

Passages that change topic (from taxes to education, say) with a new recommendation usually contain separate arguments. Passages that stay on one thesis usually contain one argument with several premises.

6.3 Worked examples

Example 1: two independent arguments

“The city should expand its metro network, since road congestion costs billions annually. Also, the city must regulate fares, because unregulated fares exclude poor commuters.”

Final conclusions: (1) expand the metro, (2) regulate fares. They do not support each other. Count: **2 arguments**, each with one premise. Figure 1 shows the map.

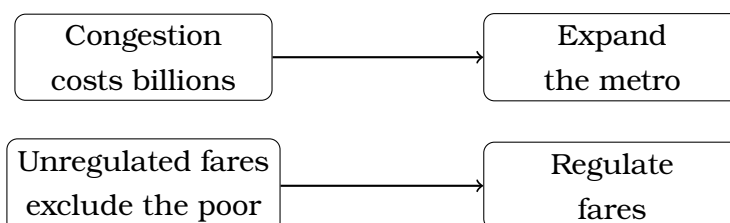


Figure 1: Two independent premise-to-conclusion pairs give two arguments.

Example 2: one chain counts as one argument

“Heavy fines for littering reduce street litter. Cleaner streets attract more tourists. More tourists raise local incomes. So the municipality should impose heavy fines.”

Support map: fines → cleaner streets → more tourists → higher incomes → the municipality should impose heavy fines. Each intermediate claim feeds the next. Only the final recommendation is a conclusion that supports nothing else. Count: **1 argument**, even though there are four linked claims.

Example 3: a rejected view does not count

“Some say that coaching centres should be banned because they commercialise education. This is mistaken: banning them would only push students to unregulated tutors. Coaching should therefore be regulated, not banned.”

The first claim is a counter-view, reported and rejected. The author’s conclusion is that coaching should be regulated, supported by the claim that a ban would push students to unregulated tutors. Count of the author’s arguments: **1**. If the stem asked for “arguments discussed”, the answer could become **2**.

Common mistakes in counting questions

- Counting premises instead of conclusions.
- Counting a chain as several arguments.
- Counting an opposing view the author rejects.
- Counting two premises that jointly support one conclusion as two arguments.

Exam-speed version

List the author’s final conclusions, the ones that support nothing further, and count only those.

7 Algorithm 4: Making Inferences

7.1 Definitions and parameters

Inference

An **inference** is a statement that must be true, or is strongly supported, given the passage alone. It is not written in the passage, but the passage forces it or strongly backs it.

There are three kinds of inference question, and the stem tells you which.

- **Strict inference** (“must be true”, “necessarily follows”): the option has no escape route. If the passage is true, the option is true.
- **Best-supported inference** (“most strongly supported”): several options may be partly supported, and you choose the one with the most backing in the text. It need not be certain.
- **Author’s view** (“the author would agree”): what the author believes, even if it is not literally stated.

7.2 Inference compared with its neighbours

Students often mix up these terms. The table fixes them.

Term	What it is	Direction
Premise	A stated reason	Passage → conclusion
Conclusion	The stated claim the premises support	Premise → it
Assumption	An <i>unstated</i> premise the argument needs	Fills the gap before the conclusion
Inference	Something that follows from what is stated	Passage → new statement
Strengtheners	A new fact that makes the conclusion more likely	Outside fact → conclusion

7.3 The four checks

For every option, run these four checks.

- (1) **Scope**: does the option stay within the topic of the passage?
- (2) **Strength**: is it equal to or weaker than the passage’s claim? (See the golden strength rule in Section 2.)

- (3) **Direction:** is the relationship the right way round?
- (4) **Traceability:** can you underline the exact words in the passage that support it?

7.4 Common invalid inferences (traps)

- **Reversal.** “All doctors are graduates” does *not* mean “All graduates are doctors”.
- **Overgeneralisation.** “Some” becomes “all” or “most”.
- **Strengthening.** “May” becomes “will”; “is linked to” becomes “causes”.
- **Outside knowledge.** The option is true in the real world but not supported by the passage.
- **Extreme words.** *Only, always, never, entirely, solely, completely*, where the passage has no such word.
- **Prediction.** The passage describes the present; the option describes the future.
- **Recommendation.** The passage describes; the option says what *should* be done.

7.5 The algorithm

- (1) **Read the stem precisely.** Is it asking for “must be true”, “best supported” or “author would agree”?
- (2) **Rewrite the passage as a short fact list**, one line per claim. Flag every quantifier and modal.
- (3) **For each option, run the four checks:** scope, strength, direction, traceability.
- (4) **Run the negation test.** Negate the option. If the passage now contradicts the negated option, the original option is valid. (For how to negate statements correctly, see Section 14.)
- (5) **Eliminate** any option that fails one check or falls into the trap list.
- (6) **If two options survive**, use this tiebreaker: prefer the narrower, more cautious one that is fully backed by the text. This is a heuristic, not a law; it works because a narrower claim has fewer ways to be wrong.

- (7) **Last check.** Can you complete the sentence “This follows because the passage says ____”? If you cannot fill the blank, reject the option.

7.6 Worked examples

Example 1: a strict inference

“Most of the district’s schools lack libraries. The district’s literacy rate is below the state average.”

Option	Verdict	Reason
Lack of libraries causes low literacy.	Invalid	Correlation turned into causation
All schools in the district lack libraries.	Invalid	“Most” cannot become “all”
At least some schools in the district lack libraries.	Valid	“Most” guarantees “some”
Other districts’ schools all have libraries.	Invalid	Outside the scope

Example 2: a best-supported inference

“Since the introduction of the helmet law, hospital admissions for head injuries among two-wheeler riders in the city have fallen by a third. Over the same period, the number of registered two-wheelers has risen.”

Options: (a) The helmet law caused the fall in admissions. (b) Fewer riders wear helmets than before. (c) Admissions for head injuries among riders fell even though two-wheeler registrations increased. (d) Head injuries among all road users fell by a third.

(a) fails strength: the passage reports a sequence in time, not a cause. (b) is the opposite of what the passage hints. (c) restates two facts the passage states, so it passes traceability, scope, strength and direction. (d) fails scope: the passage concerns *riders* and *hospital admissions*, not all road users or all injuries.

Answer: (c).

Common mistakes in inference questions

- Choosing an option because it is true in real life.
- Choosing a stronger option because it “sounds like” the passage’s theme.
- Confusing “best supported” with “must be true” and rejecting a good

option for not being certain.

Exam-speed version

For each option ask: in scope? no stronger than the passage? right direction? can I point to the words? Reject on the first “no”.

8 Algorithm 5: Identifying Assumptions

8.1 Definitions and parameters

Assumption

An **assumption** is an unstated statement that must be true for the conclusion to follow from the premises.

A **necessary assumption** is one without which the argument collapses (if it is false, the conclusion no longer follows). A **sufficient assumption** is one that, if true, guarantees the conclusion. Most exam questions ask for necessary assumptions.

The **gap** is the space between what the premises talk about and what the conclusion talks about.

The parameters you need are: (1) the premises, (2) the conclusion, and (3) the key terms that appear on one side but not the other. That is where the gap lives.

8.2 What an assumption is *not*

- It is not *stated* in the passage (otherwise it is a premise).
- It is not merely *helpful* (that is a strengthener; see Algorithm 6).
- It is not a restatement of the conclusion.
- It is not a broad claim far larger than the argument needs.

8.3 Checklist by argument type

Argument type	Typical assumptions
Causal (A causes B)	No other cause explains B ; B is not causing A ; it is not a coincidence
Analogy (A is like B)	A and B are similar in the respects that matter
Sample (some $\rightarrow$ all)	The sample is representative
Plan or policy	It is feasible, it achieves the goal, and it has no worse side effects
Prediction	Past trends will continue
Change of terms (e.g. "test scores" to "learning")	The two terms mean the same thing
Measurement	The measure used captures what it claims to

Argument type	Typical assumptions
Recommendation from facts	The goal stated is worth pursuing, and someone is responsible for acting

8.4 The algorithm

- (1) **Write the argument** as “Premise(s). Therefore, Conclusion.”
- (2) **Find the gap.** List the key terms in the premises and in the conclusion. Mark terms present on one side only.
- (3) **Identify the argument type** using the checklist above.
- (4) **Write the bridge sentence:** “For this to work, it must be true that [premise term] and [conclusion term] are connected as follows: ____.”
- (5) **Compare with the options.** Keep those that fill the gap you found.
- (6) **Apply the negation test to each surviving option** (*for necessary-assumption questions only*). Negate the option. If the argument now fails, the option is a necessary assumption. If the argument still works, reject the option. See Section 14 for how to negate correctly.
- (7) **Reject** options that are already stated in the passage, that merely help, that introduce a new topic, or that are far stronger than needed.
- (8) **Tiebreaker (heuristic):** the correct option is often modest (“at least some”, “generally”), reuses the passage’s own terms, and quietly fills the gap. Use this only after the real checks leave two options standing.

When the negation test does not apply

The negation test is a test for *necessity*. It does not suit questions that ask for something that *strengthens* or *weakens* an argument (Algorithm 6), nor questions that ask what would *guarantee* the conclusion (a sufficient assumption). For a sufficient-assumption question, ask instead: “If this option were added to the premises, would the conclusion then follow with no gap?”

8.5 Worked examples

Example 1: causal and change-of-terms

“Test scores at School A rose after it introduced a new teaching method. Therefore, the method improves learning.”

Premise: scores rose after the method. Conclusion: the method improves learning. Gaps: (a) *scores* versus *learning*; (b) *after* versus *because of*. Type: causal plus change of terms.

Assumptions: (1) test scores are a reasonable measure of learning; (2) nothing else caused the rise.

Negation check on (2): “Something else caused the rise in scores” destroys the argument, so (2) is necessary. Negation check on (1): “Test scores are no measure of learning at all” also destroys it, so (1) is necessary.

Example 2: a policy argument, and an option that fails the test

“The city should ban plastic bags. Plastic bags clog drains and cause floods.”

Gap: the premise is about what bags *do*; the conclusion is about a *ban*. The bridge: the ban would reduce the number of bags clogging drains.

Candidate (A): “A ban on plastic bags would reduce the number of plastic bags entering drains.” Negation: “A ban would *not* reduce the number entering drains.” The argument falls apart. So (A) is a necessary assumption.

Candidate (B): “Plastic bags are the *only* cause of flooding.” Negation: “Plastic bags are not the only cause of flooding.” The argument survives, since bags can still be a significant cause. So (B) is *not* an assumption; it is far too strong.

Common mistakes in assumption questions

- Choosing an option that strengthens the argument but is not required for it.
- Choosing an option already stated in the passage.
- Choosing an extreme option (*only*, *always*) that the argument does not need.
- Applying the negation test incorrectly: for example, negating “all” as “none” when the correct negation is “not all”.

Exam-speed version

Find the word in the conclusion that the premise never mentions, and the link the author takes for granted. The correct option states that link in the mildest form. Negate it: does the argument collapse?

9 Algorithm 6: Strengthening and Weakening

9.1 Definitions and parameters

Strengthen and weaken

To **strengthen** an argument is to make its conclusion *more likely* to be true. To **weaken** it is to make the conclusion *less likely*. Neither requires proving or disproving the conclusion. A strengthener does not have to be decisive, and a weakener does not have to refute.

Parameters: the conclusion, the premises, and the assumption (Algorithm 5). A strengthener or weakener works by acting on the assumption.

9.2 Ways to strengthen and weaken

Argument type	Strengthen by...	Weaken by...
Causal	Ruling out an alternative cause; showing the effect is absent when the cause is absent	Showing an alternative cause; showing the cause occurred without the effect; reversing cause and effect
Sample or survey	Showing the sample is representative	Showing the sample is unrepresentative or too small
Analogy	Showing the cases are similar in relevant respects	Showing a relevant difference between the cases
Plan or policy	Showing it is feasible, and works elsewhere	Showing it is impractical, or has harmful side effects
Prediction	Showing the trend is stable	Showing something will change the trend

9.3 The algorithm

- (1) Identify the conclusion and premises, then find the assumption (Algorithm 5).
- (2) Read the stem: strengthen or weaken? Mark it, since it is easy to answer the wrong way round.
- (3) For each option ask: “Does this make the assumption *more* or *less* likely?”
- (4) Choose the option that moves the assumption in the direction the stem requires.

- (5) Reject options that are irrelevant, that merely restate a premise, or that push the opposite way.
- (6) If two options both help, prefer the one that acts on the assumption *central* to the argument rather than on a side issue.

9.4 Worked example

Example

Argument: “Test scores at School A rose after it introduced a new teaching method. Therefore, the method improves learning.” The assumptions are that scores measure learning and that nothing else caused the rise.

Strengthen. “Nearby schools that did not adopt the method saw no rise in scores over the same period.” This rules out a general cause, such as a rise in scores everywhere, and so makes the assumption “nothing else caused it” more likely.

Weaken. “In the same year, School A replaced its examination with a noticeably easier one.” This supplies an alternative cause for the rise and so makes the assumption less likely.

Irrelevant. “School A is a large school.” Nothing links this to the assumption.

Common mistakes

- Answering the opposite question (choosing a strengthener for a weaken stem).
- Choosing an option that restates the premise, which adds nothing.
- Expecting the option to prove or disprove. It only has to shift the likelihood.

Exam-speed version

Name the gap. Strengthen = close it or block rivals. Weaken = widen it or supply a rival. Check the direction in the stem before you mark.

10 Algorithm 7: Identifying Flaws in Reasoning

10.1 Definitions and parameters

Flaw

A **flaw** is a mistake in the logic connecting premises to conclusion. A flawed argument is one whose conclusion does not follow from its premises. It does *not* mean the conclusion is false.

The fallacy fallacy

If an argument contains a flaw, you may only say the *argument* fails to prove its conclusion. The conclusion itself might still be true for other reasons. Also, several items on the usual lists are only *sometimes* flawed. Citing a genuine expert is reasonable evidence, though not proof. Questioning a witness's honesty is relevant when the witness's testimony is the evidence. The right question is always: does this step actually support the conclusion?

10.2 Common flaws

Flaw	Meaning	Example	When it is <i>not</i> a flaw
Correlation taken as causation	Two things occur together, so one is said to cause the other	Ice-cream sales and drownings both rise in summer, so ice cream causes drowning	When controlled evidence also shows a causal link
Hasty generalisation	Conclusion drawn from too few cases	Two rude waiters, so all waiters are rude	When the sample is large and representative
False dilemma	Only two options offered when more exist	Either we ban it or society collapses	When there really are only two options
Ad hominem	Attacking the person rather than the claim	He is a failure, so his plan is wrong	When the person's credibility is itself the issue, as for a witness
Appeal to authority	Truth based on who said it	A famous actor says it works	When the authority is a genuine expert in the relevant field
Appeal to popularity	True because many believe it	Everyone uses it, so it is safe	When popularity is itself the point, such as a poll of voters

Flaw	Meaning	Example	When it is <i>not</i> a flaw
Straw man	Distorting an opposing view and attacking the distortion	Critics want no rules at all	Never acceptable as a rebuttal
Circular reasoning	The conclusion is used as its own premise	It is true because it says so	Never valid as proof
Composition and division	What is true of the parts is assumed true of the whole, or the reverse	Each brick is light, so the wall is light	When the property really carries over
Equivocation	A word changes meaning mid-argument	Using “law” for both a legal rule and a scientific law	Never valid
Slippery slope	One step is claimed to lead inevitably to extremes	Allow this and soon everything is allowed	When the chain of consequences is evidenced
Affirming the consequent	Reversing a conditional	If it rains the road is wet; the road is wet, so it rained	Never valid (see Section 14)
Necessary versus sufficient confusion	Treating a requirement as a guarantee	Passing the written test is required, so anyone who passes gets the licence	Never valid
Appeal to ignorance	Claiming something is true because it is not proved false	No one has disproved it, so it is true	In law, where the burden of proof is allocated by rule

10.3 The algorithm

- (1) Extract the premise(s) and the conclusion.
- (2) Ask what step the author takes between them (generalising, assuming a cause, attacking a person, and so on).
- (3) Match that step to the table.
- (4) Choose the option that *names that error accurately*, using the vocabulary of the passage.
- (5) Check that the option describes a move the author actually made, not a flaw that merely sounds sophisticated.

Example

“Since the new traffic rules came into force, accidents in the city have fallen. The rules clearly work.”

Premise: accidents fell after the rules. Conclusion: the rules work. The step: from “after” to “because”. The flaw: taking a sequence in time as proof of causation. Possible alternative causes (better roads, fewer vehicles, a rainy season) have not been ruled out.

Exam-speed version

Strip to premise and conclusion, name the jump between them, and pick the option describing exactly that jump.

11 Algorithm 8: Main Idea

11.1 Definitions and parameters

Main idea and its neighbours

Topic: what the passage is about (a noun phrase). **Main idea:** the central point the author makes *about* the topic, supported by the whole passage.

Supporting detail: an example, statistic or sub-point that backs the main idea. **Purpose:** why the author wrote the passage (to explain, argue, warn, and so on).

11.2 The algorithm

- (1) Check the first and last sentences of the passage and of each paragraph. Authors often state or restate the main idea there.
- (2) Ask: “If I could keep only one sentence, which would it be?”
- (3) Write the main idea in your own words before looking at the options.
- (4) Classify each option:
 - **Too narrow:** covers one example or one paragraph only.
 - **Too broad:** goes beyond the passage.
 - **Distortion:** changes the strength or direction of the author’s claim.
 - **Opposite:** contradicts the passage.
 - **Half-right:** starts correct and ends wrong.
- (5) Choose the option that covers every paragraph’s contribution and nothing more.

Worked example

“Cities are rapidly losing green spaces to construction. Yet these spaces cool neighbourhoods, absorb stormwater and improve residents’ mental health. Planners who treat them as spare land therefore misjudge their value; they should be counted as infrastructure, just like roads and drains.”

Options: (a) Cities are losing green spaces. (b) Green spaces cool neighbourhoods. (c) Green spaces should be valued and planned as essential infrastructure rather than spare land. (d) Construction in cities should be stopped.

(a) is background: too narrow. (b) is one supporting detail: too narrow. (d)

is too broad and is never said. **Answer: (c).**

Exam-speed version

Write the point in one line first. Reject options that are one example (too narrow) or go beyond the passage (too broad).

12 Algorithm 9: Purpose of a Sentence or Paragraph

12.1 Definitions and parameters

Function

The **function** (or purpose) of a sentence is the *job it does for the argument*, as opposed to its *content*, which is what it says. A question about function asks “why is this sentence here?”

12.2 Common functions and their signals

Function	What the sentence does	Signals
Illustrates	Gives an example of a general point	<i>for example, consider, such as</i>
Supports	Offers evidence or a reason	<i>because, studies show</i>
Concedes	Admits a point for the sake of fairness	<i>admittedly, it is true that</i>
Introduces a counter-view	Presents an opposing position	<i>some argue, critics claim</i>
Rebuts	Answers a counter-view	<i>but, however, this ignores</i>
Defines	Explains a term	<i>means, refers to, is defined as</i>
Concludes	States the final point	<i>therefore, thus, hence</i>
Qualifies	Limits the scope of a claim	<i>usually, in most cases, unless</i>
Transitions	Moves from one part to the next	<i>turning to, next</i>

12.3 The algorithm

- (1) Locate the sentence and read the sentences before and after it.
- (2) Ask what job it does: illustrate, concede, rebut, define, conclude, or another.
- (3) Match that job to the option's *verb* (*illustrates, qualifies, refutes, summarises*).
- (4) Ignore options that describe the content but not the function.
- (5) Check that the job fits the sentence's position (an illustration usually follows the general point).

Worked example

In a passage arguing that remote work raises productivity, the sentence “*Consider the Pune call centre that cut its attrition by half after allowing home working*” has the function of **illustrating** the general claim by a specific case. An option such as “it presents a counter-view” is wrong because nothing in the sentence opposes the claim.

Exam-speed version

Ask “why is this here?” and match the answer’s verb: illustrates, concedes, rebuts, defines, concludes.

13 Algorithm 10: Resolving a Paradox

13.1 Definitions and parameters

Paradox question

A **paradox** in these questions is a pair of facts that appear to contradict each other. The task is to find the option that, if true, shows how *both* facts can be true together. You are explaining, not choosing a side.

13.2 Typical resolutions

- **Different groups:** the two facts describe different groups of people or things.
- **Different times:** the facts concern different periods.
- **Different measures:** the two facts use different measures (total versus per head, say).
- **Hidden factor:** a third factor pushes in the opposite direction.

13.3 The algorithm

- (1) State the two apparently conflicting facts in your own words.
- (2) Ask: “What single new fact would make both true at once?”
- (3) Test each option: does it explain *both* sides?
- (4) Reject options that explain only one side, that deepen the conflict, or that deny one of the facts.

Worked example

“A city introduced a congestion charge, and traffic inside the charged zone fell sharply. Yet average travel times across the city rose.”

Conflict: less traffic in the zone, yet slower trips across the city. Resolution: “Many drivers diverted onto ring roads that were already congested.” This is a hidden factor that explains how total travel times can rise while the zone itself clears. An option like “The charge raised revenue for the city” explains neither fact and is irrelevant.

Exam-speed version

Write the two facts. The answer is the new fact that makes both true, usually a different group, time, measure or hidden factor.

14 Algorithm 11: Conditional Logic

14.1 Definitions and parameters

Conditional statements

In “If P then Q ” (written $P \rightarrow Q$), P is the **sufficient condition** (having P is enough to guarantee Q) and Q is the **necessary condition** (you cannot have P without Q). The **contrapositive** of $P \rightarrow Q$ is $\neg Q \rightarrow \neg P$ (“if not Q then not P ”), and it is logically equivalent to the original. The **converse** is $Q \rightarrow P$. The **inverse** is $\neg P \rightarrow \neg Q$. Neither the converse nor the inverse follows from the original.

14.2 Translation table: turning English into $P \rightarrow Q$

English	Logical form	Contrapositive
If P then Q	$P \rightarrow Q$	$\neg Q \rightarrow \neg P$
P only if Q	$P \rightarrow Q$	$\neg Q \rightarrow \neg P$
P if Q	$Q \rightarrow P$	$\neg P \rightarrow \neg Q$
Only P are Q	$Q \rightarrow P$	$\neg P \rightarrow \neg Q$
All P are Q	$P \rightarrow Q$	$\neg Q \rightarrow \neg P$
No P are Q	$P \rightarrow \neg Q$	$Q \rightarrow \neg P$
P unless Q	$\neg Q \rightarrow P$	$\neg P \rightarrow Q$
P is necessary for Q	$Q \rightarrow P$	$\neg P \rightarrow \neg Q$
P is sufficient for Q	$P \rightarrow Q$	$\neg Q \rightarrow \neg P$

14.3 Valid and invalid moves

Given	Valid	Invalid
$P \rightarrow Q$ and P	Q	
$P \rightarrow Q$ and $\neg Q$	$\neg P$	
$P \rightarrow Q$ and Q	nothing follows about P	P (affirming the consequent)
$P \rightarrow Q$ and $\neg P$	nothing follows about Q	$\neg Q$ (denying the antecedent)
$P \rightarrow Q$ and $Q \rightarrow R$	$P \rightarrow R$	$R \rightarrow P$
P or Q (at least one) and $\neg P$	Q	

On “unless” and on “all”

“You will fail unless you study” strictly gives only: if you do not study, you fail ($\neg S \rightarrow F$). It does not strictly say you pass if you study. Everyday speech often treats it as two-way, but exam logic does not.

“All P are Q ” is usually taken to imply “Some Q are P ”. That step is valid only if there is at least one P (if P is an empty group, “all P are Q ” is true but “some Q are P ” is not). Exams normally treat the step as valid, but know the hidden assumption.

14.4 Negating statements correctly

Wrong negations cause many errors in the negation test (Algorithms 4 and 5). Memorise these.

Statement	Its negation
All A are B	Some A are not B (not “No A are B ”)
Some A are B	No A are B
No A are B	Some A are B
Always	Sometimes not
Never	Sometimes
P and Q	Not P , or not Q (or both)
P or Q	Neither P nor Q
If P then Q	P and not Q
It is possible that P	P is impossible

14.5 The algorithm

- (1) Translate every statement into $P \rightarrow Q$ form using the translation table.
- (2) Write its contrapositive.
- (3) Chain conditionals where the consequent of one is the antecedent of the next.
- (4) Test each option against the original, the contrapositive and the chains only. Reject anything that matches the converse or inverse.

Worked example

“Only registered voters may vote. Meera did not vote.”

“Only registered voters may vote” becomes: if a person votes, then the person is registered ($V \rightarrow R$). Contrapositive: if not registered, then did not vote. Meera did not vote ($\neg V$). Is she unregistered? From $V \rightarrow R$ and $\neg V$,

nothing follows about R (denying the antecedent). Meera may be registered and have chosen not to vote. A conclusion that “Meera is not registered” is therefore **invalid**.

Exam-speed version

Convert to “if ... then”. Write the contrapositive. Accept only the original, its contrapositive, and chains. Reject the converse and inverse.

15 Algorithm 12: Syllogisms and Venn Diagrams

Check the official bulletin to see whether your target exam uses syllogism-style questions. The method below also helps in any question with “all”, “some” and “no”.

15.1 Definitions and parameters

Syllogism

A **syllogism** is an argument built from statements about groups, with two premises and a conclusion. A **categorical statement** comes in four standard forms:

- **All A are B** (universal affirmative)
- **No A are B** (universal negative)
- **Some A are B** (particular affirmative)
- **Some A are not B** (particular negative)

A conclusion **definitely follows** if it is true in *every* diagram that fits the premises. It is only **possible** if it is true in *some* diagrams. Do not confuse the two.

15.2 How to draw each statement

- **All A are B** : draw circle A completely inside circle B .
- **No A are B** : draw the two circles apart, with no overlap.
- **Some A are B** : draw the circles overlapping and place a mark in the overlap.
- **Some A are not B** : draw a mark in A outside B .

15.3 The algorithm

- (1) Draw the minimum diagram that makes both premises true.
- (2) Ask which parts are still undecided (where the diagram could be drawn differently without breaking the premises). Draw the alternatives.
- (3) Test the conclusion. If it holds in every legitimate diagram, it definitely follows. If it holds in some but not all, it is only possible. If it fails in some, it does not follow.

- (4) For “some” conclusions, you need a mark guaranteed in the overlap in all diagrams.

15.4 Standard results you can rely on

Premises	Valid conclusion
All A are B ; all B are C	All A are C
All A are B ; no B are C	No A are C
Some A are B ; all B are C	Some A are C
Some A are B ; no B are C	Some A are not C
No A are B ; all C are A	No C are B
All A are B ; some B are C	No definite conclusion about A and C

Worked example

“All doctors are graduates. Some graduates are artists.” Does “Some doctors are artists” follow?

Draw doctors inside graduates. The artist circle must overlap graduates, but nothing says whether it overlaps the doctors’ circle. Figure 2 shows two legitimate diagrams. In the left one, no doctor is an artist, so the conclusion is false. In the right one, it is true. It is true in only some diagrams, so it is **possible but does not definitely follow**.

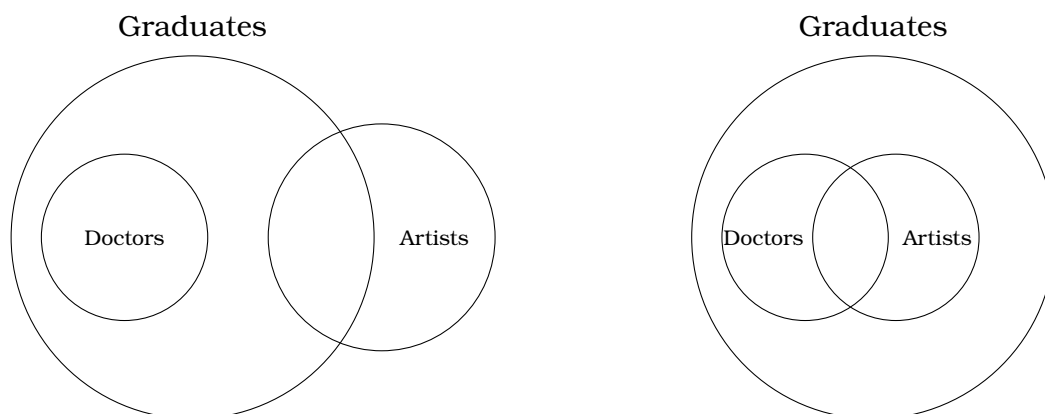


Figure 2: Two diagrams consistent with the premises. In the left, no doctor is an artist; in the right, some are.

Common mistakes

- Treating “possible” as “definite”.
- Drawing only one diagram when the premises allow several.
- Forgetting that “some” does not exclude “all”.
- Using real-world knowledge (about whether doctors are artists) rather than the stated premises.

Exam-speed version

Draw it. Redraw it in every legitimate way. If the conclusion holds in all drawings, it follows; if only some, it is merely possible.

16 Algorithm 13: Legal Reasoning

16.1 What these questions are

Legal Reasoning questions are built from passages that state rules, principles or legal developments and then ask you to apply or interpret them. Skills tested include applying a stated principle to a set of facts, spotting which reasoning is correct, and (in some questions) basic legal knowledge. You do not need to be a lawyer, but you do need a small working vocabulary.

16.2 A beginner's legal glossary

Term	Meaning
Plaintiff, claimant	The person who sues in a civil case
Defendant	The person who is sued (civil) or accused (criminal)
Tort	A civil wrong, other than breach of contract, for which the law gives a remedy, usually damages
Negligence	Failing to take reasonable care towards someone to whom you owe a duty, causing them loss
Contract	An agreement that the law will enforce
Offer and acceptance	The proposal made by one party and the agreement to it by the other
Consideration	Something of value given in exchange for a promise
Void	Having no legal effect from the start
Voidable	Valid until the party entitled to do so chooses to cancel it
Liability	Legal responsibility to pay or make good a loss, or to be punished
Damages	Money awarded as compensation
Vicarious liability	Liability of one person for the wrong of another, such as an employer for an employee's act in the course of employment
Strict liability	Liability without proof of fault
Mens rea	The guilty mind: the mental element of a crime
Actus reus	The guilty act: the physical element of a crime
Natural justice	Basic fairness in decision-making, including hearing the other side and having an unbiased decision-maker
Precedent	An earlier decision that guides later cases

Common legal maxims.

Maxim	Meaning
<i>Audi alteram partem</i>	Hear the other side
<i>Nemo iudex in causa sua</i>	No one should be a judge in their own cause

Maxim	Meaning
<i>Actus non facit reum nisi mens sit rea</i>	An act is not guilty unless the mind is guilty
<i>Ignorantia juris non excusat</i>	Ignorance of the law is no excuse
<i>Volenti non fit injuria</i>	No wrong is done to one who consents
<i>Res ipsa loquitur</i>	The thing speaks for itself
<i>Caveat emptor</i>	Let the buyer beware
<i>Ubi jus ibi remedium</i>	Where there is a right, there is a remedy

16.3 Three kinds of question

- (1) **Apply the principle to the facts:** “Is X liable?”, “What is the likely outcome?”
- (2) **Identify the correct reasoning:** the outcome may be right in several options, but only one gives the right *reason*.
- (3) **Legal knowledge:** questions that rely on a legal concept or maxim.

16.4 The algorithm for applying a principle to facts

- (1) **Read the stem first** so you know exactly what is asked (liability of whom? which reasoning is correct?).
- (2) **Write the principle in one line.**
- (3) **Break it into elements.** For example, a negligence principle might need: a duty of care + breach + damage caused by the breach.
- (4) **List the material facts.** Drop irrelevant details (names, places, amounts, unless the principle depends on them).
- (5) **Test each element against the facts, one by one.**
- (6) **Check for exceptions and defences** stated in the principle. A defence can remove liability even when every element is satisfied.
- (7) **Check what the question tells you about real law.** If it says to ignore real law (or to apply only the stated principle), apply only the stated principle, even if you disagree with it. If it says nothing, use only what is stated and do not import outside doctrine unless a legal-knowledge question requires it.
- (8) **Choose the option whose result and reasoning match your analysis.** An option with the right outcome and the wrong reason is wrong.

16.5 Worked examples

Example 1: elements

Principle. A person who, by failing to take reasonable care, causes damage to another to whom he owes a duty of care is liable to pay damages. A duty of care is owed to people who can reasonably be foreseen to be affected.

Facts. Ravi, while texting, drives through a red light and hits Meena's parked car, damaging it.

Elements: (1) duty of care: other road users and their property are foreseeably affected; satisfied. (2) failure to take reasonable care: texting while driving through a red light; satisfied. (3) damage caused by the failure: the car was damaged by the collision; satisfied. No exception is stated. **Ravi is liable.**

Example 2: an exception

Principle. As in Example 1, with one addition: no liability arises if the damage was caused by a sudden medical emergency which the person could not have foreseen.

Facts. Ravi, who has never had any health problem, suffers a sudden seizure, loses control and hits Meena's parked car.

The elements look met, but the exception applies: the cause was a sudden, unforeseeable medical emergency. **Ravi is not liable.** Notice that sympathy for Meena's loss is irrelevant to the test.

Example 3: a principle that differs from real law

Principle. An offer, once made, cannot be withdrawn for seven days from the day it is made.

Facts. On Monday, Anil offers to sell his bicycle to Bina for Rs. 5,000. On Wednesday he writes to withdraw the offer. On Thursday, Bina accepts.

In ordinary law an offer can generally be withdrawn before it is accepted. But the question gives a different principle, so apply it: Anil's withdrawal on Wednesday is within the seven-day period and is therefore ineffective, and Bina's acceptance on Thursday is within the period. A binding agreement exists.

Common mistakes in legal reasoning

- Importing real law when the question says to apply only the stated principle.

- Adding facts the passage does not give.
- Letting sympathy or fairness decide the outcome.
- Missing an exception or defence stated in the principle.
- Choosing an option with the right outcome but the wrong reasoning.
- Confusing motive (why someone acted) with intention (what they meant to do), when the principle requires the latter.

Exam-speed version

Principle in one line. Elements as a checklist. Tick each against the facts. Look for the exception. Check the reasoning, not just the result.

17 Algorithm 14: Vocabulary in Context

17.1 Definitions and parameters

Key terms

Denotation: a word's dictionary meaning. **Connotation:** the feelings or associations the word carries. **Polysemy:** a single word having several meanings (*sound* can mean healthy, valid, or a noise). **Idiom:** a fixed phrase whose meaning is not the sum of its words (*break the ice*). **Context:** the surrounding words and sentences that fix which meaning is intended.

17.2 The algorithm

- (1) Cover the word and read the sentence.
- (2) Predict a replacement word or phrase from the context alone.
- (3) Match your prediction to the options.
- (4) Test the winner: substitute it into the sentence and check that it reads naturally and keeps the sentence's tone.
- (5) Do not use the word's most common meaning if it clashes with the sentence.
- (6) If you do not know the word, break it into parts (prefix, root, suffix) and use the polarity of the sentence (positive or negative) to eliminate options.

17.3 Useful word parts

Part	Meaning	Example
<i>mal-</i>	bad	malevolent, malfunction
<i>bene-</i>	good, well	benefit, benevolent
<i>anti-, contra-</i>	against	antidote, contradict
<i>circum-</i>	around	circumvent, circumference
<i>pre-, post-</i>	before, after	precedent, postpone
<i>dict</i>	say	predict, verdict
<i>scrib, script</i>	write	describe, manuscript
<i>spec, spect</i>	look	inspect, spectator
<i>port</i>	carry	transport, import
<i>jud, jur</i>	law, right	judge, jury
<i>-cide</i>	killing	suicide, pesticide
<i>-phobia</i>	fear	claustrophobia

Worked example

“The court found the petitioner’s claim to be sound and allowed the appeal.”

Cover *sound*. A court that *allows* an appeal after assessing a *claim* has found the claim valid or well-reasoned. *Sound* here does not mean a noise or physically healthy. Predicted meaning: *valid*. Choose the option that means valid or well-founded.

Exam-speed version

Cover the word, predict, match, then test by substituting.

18 Algorithm 15: Parallel Reasoning

18.1 Definitions and parameters

Parallel reasoning

A **parallel reasoning** question gives you an argument and asks you to choose the option whose *logical structure* is the same, regardless of topic. If the original argument contains a flaw, the parallel argument must contain the *same* flaw.

18.2 The algorithm

- (1) Reduce the original to symbols (for example, “All A are B ; C is A ; so C is B ”).
- (2) Preserve the strength words exactly: all, some, most, may, must.
- (3) Reduce each option to symbols in the same way.
- (4) Choose the option with an identical structure, with the same strengths and the same direction.
- (5) Check that the option does not reverse a conditional or change “all” to “some”.

Worked example

Original: “All metals conduct electricity; copper is a metal; so copper conducts electricity.” Structure: All A are B ; c is A ; so c is B .

Option	Verdict
All birds lay eggs; a sparrow is a bird; so a sparrow lays eggs.	Parallel
Some fruits are sour; lemons are fruits; so lemons are sour.	Not parallel (“some” in the first premise)
All metals conduct electricity; copper conducts electricity; so copper is a metal.	Not parallel (the direction is reversed)

Exam-speed version

Symbols, then strengths, then direction. The match must have all three the same.

19 Algorithm 16: Universal Option Elimination

19.1 Why this algorithm matters

Every multiple-choice question can be attacked by eliminating wrong options. Most wrong options belong to a small number of types. If you can *name* the flaw in each wrong option, you answer faster, and when you make a mistake you know which trap caught you.

19.2 The wrong-answer types

Type	What it is	How to spot it
Out of scope	Introduces something the passage never mentions	You cannot find its key term in the passage
Extreme	Uses <i>only</i> , <i>always</i> , <i>never</i> , <i>all</i> , <i>entirely</i> without support	Look for absolute words
Reversal	Gets the direction of a relationship backwards	Check which term causes or contains which
Half-right	Starts correctly and ends wrongly	The second half fails the check
True but irrelevant	Correct in real life or in the passage, but does not answer the question asked	Re-read the stem, then ask “does this answer it?”
Opposite	Contradicts the passage	Compare with the exact words
Distortion	Changes the strength or tone of the author’s claim	Compare quantifiers and modals
Wrong speaker	Attributes someone else’s view to the author	Check who said it
Too narrow or too broad	Covers one detail only, or goes beyond the passage	Compare with the main idea

19.3 The algorithm

- (1) **Predict before you read the options.** For most questions you can state the answer, or at least its shape, from the passage.
- (2) **Label every option** with one of the types above, or with “possible”.
- (3) **Compare the survivors against the exact words of the stem** (“must be true”, “best supported”, “weakens”).
- (4) **Tiebreakers (heuristics, not laws):** prefer the option that uses the passage’s own terms, the narrower claim, and the milder strength, provided each is fully backed by the text.

- (5) **Final check:** the passage would have to be wrong for the answer to be wrong. If the passage could be true while your answer is false, reconsider.

Exam-speed version

Predict. Label each wrong option by type. Compare finalists to the stem.
Pick the one the passage backs with its own words.

20 Algorithm 17: Time Triage and Guessing

20.1 Definitions and parameters

Key terms

Triage: deciding quickly which questions to do now, which to do later, and which to abandon. **Attempt rate:** the number of questions attempted divided by the number available. **Net score:** correct answers minus the penalty for wrong ones. **Expected value (EV):** the average score you would get from a decision if you repeated it many times.

20.2 Time budget

With 120 questions in 120 minutes, the average is one minute per question, including reading. A passage of about 450 words takes roughly two minutes to read at an ordinary reading speed. A rough budget is below. Adjust it from your own timed practice, because it is only a starting point.

Section	Rough budget
English Language	about 1 minute per question
Current Affairs and GK	less than 1 minute per question (you either know it or you do not)
Legal Reasoning	about 1 to 1.25 minutes per question
Logical Reasoning	about 1 to 1.25 minutes per question
Quantitative Techniques	flexible, with a cap of about 2 minutes per question
Reserve	5 to 8 minutes for review and marking the OMR sheet

20.3 The guessing rule

Suppose a correct answer earns +1 and a wrong answer costs q marks. If your probability of being right is p , the expected value of guessing is

$$EV = p \cdot 1 - (1 - p) q.$$

Guessing has positive expected value when $p > \frac{q}{1 + q}$. For $q = 0.25$ (CLAT), the break-even probability is $0.25/1.25 = 0.20$, that is 20 per cent.

Options left after elimination	Chance	EV with $q = 0.25$
4 (blind guess)	25%	$0.25 - 0.1875 = +0.0625$
3	33.3%	$0.333 - 0.1667 = +0.167$
2	50%	$0.50 - 0.125 = +0.375$

A caution on this arithmetic

These numbers assume your eliminations are *reliable*. If you sometimes eliminate the correct option by mistake, your real chance is lower than the table says. A blind guess has only a tiny positive expected value, so the benefit is real but small, and the variance (the swing in your score) is larger than the benefit. Use the table as a reason to attempt questions where you can genuinely eliminate options, not as a licence to guess wildly. For AILET or any exam with a different penalty, recompute using the formula above with the actual q .

20.4 The algorithm

- (1) Before the exam, decide your section order from timed practice (strongest first, or the order that makes you calmest). Confirm that the instructions allow free movement between sections.
- (2) Skim each passage's questions and decide: attempt now, mark for later, or skip.
- (3) Set a hard cap of about 90 seconds on a stuck question. Eliminate what you can, choose, mark it, and move on.
- (4) Attempt questions where you can reduce the options to two or three. Leave a question blank only when you cannot eliminate anything and are under severe time pressure.
- (5) Transfer answers to the OMR sheet at a fixed rhythm (for example, at the end of each section), and keep the reserve minutes for this.
- (6) In the last minutes, return to marked questions in order of how close you were to an answer.

Exam-speed version

One minute average. Cap a stuck question at about 90 seconds. Guess only after eliminating at least one option. Keep a reserve for the OMR sheet.

21 Algorithm 18: Quantitative Techniques and Data Interpretation

21.1 Definitions and parameters

This section carries about 10 to 14 questions at roughly Class 10 level. Typical topics are percentages, ratios, averages, profit and loss, simple and compound interest, speed and time, work and time, basic mensuration, basic algebra, simple probability and data interpretation (DI), which means reading tables and charts to answer questions.

Terms that cause errors

Percentage change: the change divided by the *original* value, times 100.

Percentage point: the plain difference between two percentages. **Base:** the quantity a percentage is taken of. **Weighted average:** an average in which some values count more than others.

21.2 Formulas to know

Topic	Formula
Percentage change	$\frac{\text{new} - \text{old}}{\text{old}} \times 100$
Successive changes	Multiply the factors: +20% then -20% gives $1.2 \times 0.8 = 0.96$, a net fall of 4%
Average	$\frac{\text{sum of values}}{\text{number of values}}$
Profit per cent	$\frac{\text{profit}}{\text{cost price}} \times 100$
Selling price	$\text{CP} \times \left(1 + \frac{p}{100}\right)$ for a profit of $p\%$
Simple interest	$\frac{P \times R \times T}{100}$
Compound amount	$P \left(1 + \frac{R}{100}\right)^n$
Speed	$\frac{\text{distance}}{\text{time}}$
Average speed, equal distances at speeds a and b	$\frac{2ab}{a + b}$
Work	Rate = $1/\text{time}$; combined rate is the sum of rates
Probability	$\frac{\text{favourable outcomes}}{\text{total outcomes}}$

Topic	Formula
Area of a circle	πr^2
Volume of a cylinder	$\pi r^2 h$

21.3 The algorithm

- (1) **Read the question before the table.** Know what you need.
- (2) **Check units and scale** (thousands, lakhs, per cent, per head).
- (3) **Identify the base** for any percentage. “Of what?”
- (4) **Estimate first.** Round the numbers and compute roughly. Often two options can be eliminated at once because the estimate rules them out.
- (5) **Calculate only as precisely as the options require.** If the options are far apart, a rough answer will do.
- (6) **Sanity-check** the result: does the size make sense? Is the sign right?
- (7) **Skip if stuck.** Quant is a good place to apply the 90-second cap.

21.4 Worked examples

Example 1: percentage change, and a trap

A company’s sales (in thousand units) were: 2022: 80; 2023: 100; 2024: 90.

Growth from 2022 to 2023: $(100 - 80)/80 \times 100 = 25\%$. Change from 2023 to 2024: $(90 - 100)/100 \times 100 = -10\%$. Overall change from 2022 to 2024: $(90 - 80)/80 \times 100 = +12.5\%$. A common error is to add 25% and -10% to get $+15\%$. That is wrong because the two percentages have different bases (80 and 100).

Example 2: percentage points

A party’s vote share rises from 20% to 25%. The difference is **5 percentage points**. The relative increase is $5/20 \times 100 = 25\%$. A question that says “by what percentage did the share increase?” wants 25%; one that says “by how many percentage points?” wants 5.

Example 3: average speed and work

A car goes a fixed distance at 40 km/h and returns at 60 km/h. The average speed is $2 \times 40 \times 60 / (40 + 60) = 48$ km/h, not 50. Separately, if A can finish a job in 6 days and B in 12 days, their combined rate is $1/6 + 1/12 = 3/12 = 1/4$ of the job per day, so together they take 4 days.

Common mistakes

- Adding percentages with different bases.
- Confusing percentage with percentage points.
- Averaging speeds directly when distances are equal.
- Ignoring units (thousands versus lakhs).
- Doing exact arithmetic when an estimate would eliminate the options.

Exam-speed version

Question first, units, base, estimate, eliminate, then compute only if two options remain.

22 Algorithm 19: Current Affairs and General Knowledge

22.1 Definitions and parameters

Current Affairs and GK carry about 28 to 32 questions, which makes them one of the largest sections. The questions are passage-based, but they usually reward knowing the underlying facts, names, institutions and ideas. The window is generally the previous year or so, so check the notification.

Key terms

Static knowledge: facts that rarely change (constitutional articles, geography, history). **Dynamic knowledge:** facts that change (appointments, judgments, treaties). **Linking:** connecting a news item to a static concept so you remember both.

22.2 The algorithm

- (1) **Daily (about 10 to 15 minutes).** Read one reliable current-affairs digest. For each important item, write one line: *who, what, where, when, why it matters*.
- (2) **Link it to a static concept.** For example, a new Supreme Court judgment on free speech links to Article 19; a new Act links to the legislative process; a treaty links to the countries and institutions involved.
- (3) **Weekly (about 30 minutes).** Consolidate the week's items into a one-page summary by category: national, international, economy, environment, science and technology, sports, awards, and legal developments.
- (4) **Monthly.** Revise the weekly pages and quiz yourself on names, numbers and dates.
- (5) **Watch precision.** Examiners test exact names, designations, numbers and the "first", "latest" or "largest" of something.
- (6) **Practise from the passage.** Many questions can be answered by reading the passage carefully; do not neglect passage skills for pure memory.

Sources

Reliable, free starting points include PIB (pib.gov.in), PRS India (prsindia.org), Supreme Court Observer (scobserver.in), LiveLaw, and the editorial

and explained sections of The Hindu and The Indian Express. Use more than one source for contested topics, and prefer original documents (a judgment, a bill, a press release) over summaries when you can.

Exam-speed version

One line per item (who, what, why it matters), one static link per item, and a weekly consolidation page.

23 Algorithm 20: The Review Loop

23.1 Why this matters most

Practising without review wastes most of the value of practice. The review loop turns every mistake into a rule you will not repeat. If you only adopt one algorithm from this manual, adopt this one.

23.2 Definitions and parameters

Key terms

Error log: a written record of each mistake with its cause. **Error type:** the category of the cause (for example, reading, concept, trap, time, careless). **Accuracy:** correct answers divided by attempted questions. **Right for the wrong reason:** a correct answer reached by a faulty method or a lucky guess. It must be reviewed like a mistake.

23.3 Error types

Type	Meaning
Reading	You misread the passage or the question stem
Concept	You did not know or misapplied a rule (for example, the contrapositive)
Trap	You fell for a wrong-answer type from Algorithm 16
Time	You rushed, or ran out of time
Careless	You knew the method but made a slip, such as bubbling the wrong option
Knowledge gap	A current-affairs fact or legal term you did not know

23.4 The algorithm for each wrong or guessed answer

- (1) **Re-solve it blind**, without looking at the key. Does the same wrong answer attract you?
- (2) **Find the exact sentence** in the passage that decides the question.
- (3) **Identify the step that failed** (for example, “Inference, strength check”).
- (4) **Label the trap type** (Algorithm 16) and the error type (the table above).
- (5) **Write a one-line rule** for the future (“*Some* never becomes *most*”).

- (6) **Re-attempt the question** after about a week, and again after about a month.

23.5 Weekly analysis

- (1) Count your errors by type and by question category.
- (2) Your top two categories are your drill targets for the coming week.
- (3) Track accuracy, attempt rate and net score for each section, not just the overall score.
- (4) Compare your timing with the budget in Algorithm 17 and adjust.

Exam-speed version (for practice, not the exam)

Re-solve blind. Find the deciding sentence. Name the failed step and the trap. Write the rule. Retry in a week.

24 Reasoning Beyond the Exam

24.1 Exam reasoning versus real-life reasoning

The algorithms in this manual are designed for a *closed-world* game: the passage is the whole truth, and outside knowledge is forbidden. Real life is *open-world*: evidence is missing, sources are unreliable, and outside knowledge usually matters. Some skills transfer well and some do not.

Feature	In the exam	In life
Source of truth	The passage only	Many sources, of varying quality
Outside knowledge	Forbidden	Essential
Missing information	Treat as unknown	Go and find it
Safest answer	The narrowest supported claim	Depends; look at stakes and evidence
Goal	The best option from four	The best decision under uncertainty

What transfers well:

- Separating claims from evidence.
- Spotting hidden assumptions.
- Checking correlation against causation.
- Noticing when “some” quietly becomes “all”.
- Distinguishing a flawed argument from a false conclusion.

What does not transfer: “ignore outside knowledge”, “choose the narrowest option”, and “treat the passage as true”.

24.2 Six tools for real life

1. Steelmanning

Before criticising a view, state it so well that its holders would say, “Yes, that is what we think.” A common procedure (often attributed to the philosopher Anatol Rapoport and popularised by Daniel Dennett) is: (a) restate the other side’s position clearly and fairly; (b) list any points of agreement; (c) mention anything you learned from them; (d) only then offer your criticism.

The opposite of a steelman is a straw man (Algorithm 7).

2. Lateral reading and the SIFT method

When you meet a claim online, do not just read the page more deeply. Leave the page and check what *other* sources say about it. The SIFT method (developed by the digital-literacy researcher Mike Caulfield) has four moves: **S**top (pause before sharing or believing); **I**ntestigate the source (who is behind this, and what is their track record?); **F**ind better coverage (what do other reliable sources say?); **T**race claims, quotes and media back to their original context.

3. Base rates and calibration

A **base rate** is how common something is in general. Ignoring it is a classic mistake.

Example. A disease affects 1 in 1,000 people. A test detects 99% of people who have it, but also wrongly flags 1% of healthy people. You test positive. How likely is it that you have the disease? Imagine 100,000 people: 100 have the disease and 99 of them test positive; 99,900 are healthy and about 999 of them test positive. Among all 1,098 positives, only 99 are truly ill, so the chance is $99/1098 \approx 9\%$, not 99%. A positive result is more worrying, but far from certain.

Calibration means that your confidence matches how often you turn out to be right. Practise by attaching rough probabilities to your beliefs, and check them later.

4. “Compared to what?”

Whenever you hear that something is large, small, rising or effective, ask: compared to what baseline, over what period, and out of how many? A 50% increase in a very rare event may still be tiny in absolute terms.

5. The pre-mortem

Before committing to a plan, imagine it has already failed, and write down why. This surfaces risks that optimism hides. It is the real-life version of looking for the assumption that could be false.

6. Second-order effects

Ask not only “What happens next?” but “And then what?” A ban may reduce one harm and, through substitution, create another (compare the plastic-bag example in Algorithm 5).

A last warning

Fallacy-spotting can be misused as a weapon in arguments. Calling something a fallacy does not by itself win the argument, and the person you are arguing with is often partly right. Treat these tools as a way of understanding more clearly, not of defeating others.

25 The Exam-Speed Cheat Sheet

This section compresses every algorithm into a few lines. Practise the long versions first, then drill these until they run silently.

Question type	Three-line check
Pre-step	Stems first; circle <i>but/therefore</i> ; five-word gist per paragraph; one-line main point
Tone	Target and speaker; polarity and intensity from loaded words (weight the last sentence); function. Pick the option backed by two words from the text
Premise	Find the claim the author most wants accepted; ask “because?”; ignore background, rejected views and pure illustrations
Counting arguments	Follow the stem’s wording; list final conclusions that support nothing further; count those
Inference	In scope? No stronger than the passage? Right direction? Can I point to the words? Reject on the first “no”
Assumption	Find the term in the conclusion missing from the premises; state the link in its mildest form; negate it to test necessity
Strengthen or weaken	Name the gap; strengthen = close it or block rivals; weaken = widen it or supply a rival; check the stem’s direction
Flaw	Strip to premise and conclusion; name the jump; pick the option that describes exactly that jump
Main idea	Write it first; reject too narrow (one example), too broad (beyond the passage) and distortions
Purpose of a sentence	Ask “why is this here?”; match the verb (illustrates, concedes, rebuts, defines, concludes)
Paradox	Write both facts; the answer is the new fact (different group, time, measure or hidden factor) that makes both true
Conditional logic	Convert to “if . . . then”; contrapositive; chains; reject converse and inverse
Syllogism	Draw; redraw all legitimate ways; definite only if true in every drawing
Legal reasoning	Principle in one line; elements as a checklist; tick against facts; look for exceptions; check reasoning as well as outcome
Vocabulary in context	Cover, predict, match, substitute
Parallel reasoning	Symbols; strengths; direction. All three must match

Question type	Three-line check
Option elimination	Predict; label each wrong option by type; compare finalists to the stem
Time and guessing	One-minute average; 90-second cap; guess after eliminating at least one; reserve time for the OMR
Quantitative	Question first; units; base; estimate; eliminate; compute only if needed
Current affairs	One line per item (who, what, why it matters); one static link; weekly page
Review loop	Re-solve blind; deciding sentence; failed step and trap; one-line rule; retry in a week

26 A Training Plan

Your target date and your starting level are unknown to me, so treat this as a template to stretch or shrink.

26.1 Three phases

- (1) **Phase 1: learn slowly (about 4 weeks).** Read this manual section by section. For each algorithm, do five to ten practice items on paper, writing out every step. Start the error log on day one.
- (2) **Phase 2: compress (about 4 weeks).** Move to the exam-speed versions. Work on whole passages with their question sets and time yourself loosely. Review every wrong or guessed answer using Algorithm 20.
- (3) **Phase 3: simulate (remaining time).** Take full-length timed papers at least once a week, in exam conditions, followed by a long review. Use the official past papers where you can.

26.2 A sample weekly template

Day	Focus
Monday	Logical reasoning passages (two algorithms in focus), plus the daily editorial
Tuesday	Legal reasoning sets, plus legal glossary and maxims
Wednesday	English passages (tone, main idea, vocabulary), plus quantitative practice
Thursday	Logical and legal mixed set, then review
Friday	Quantitative techniques and data interpretation, plus current affairs consolidation
Saturday	Full or half-length timed paper
Sunday	Long review using the error log; light reading; plan the next week

26.3 Reading practice

Your reading habit supports all of the above. Rotate subjects deliberately (law, economics, science, environment, ethics, history, literature, international affairs), because passages are drawn from all of them. While reading, predict the author's conclusion before finishing, and afterwards ask what the passage would say about a new situation. That is inference practice. Time yourself on passages of 450 to 600 words so that your reading speed matches the exam.

26.4 Checkpoints

- (1) After about 30 to 40 passages, count your errors by type. Your top two types tell you which algorithm to drill.
- (2) Check whether your accuracy is above about 80 per cent on attempted questions in each section before you push for more speed.
- (3) Review your plan every two weeks. If something is not working, change it, since no plan, including this one, is proven optimal for everyone.

A Error Log Template

Copy this table into a notebook or spreadsheet. Fill one row per wrong or guessed answer.

Date	Question type	Failed step	Trap or error type	Rule to remember	Retry on

B Official Papers and Free Resources

Check each site yourself, since access rules and pages change.

B.1 Official sources

- CLAT: consortiumofnlus.ac.in, for notifications, syllabus and past papers.
- AILET: nludelhi.ac.in, for the bulletin, pattern and past papers.

B.2 Free reading sources

- **Books and classics:** Project Gutenberg, Standard Ebooks, Wikisource, the Internet Archive.
- **Explainers and journalism:** The Conversation, Global Voices, Mongabay, Our World in Data, Aeon, Psyche.
- **Law and polity:** Constitution of India and bare Acts (indiacode.nic.in, legislative.gov.in), Indian Kanoon, Supreme Court Observer, PRS India, PIB.
- **Philosophy for dense argument practice:** the Stanford Encyclopedia of Philosophy and the Internet Encyclopedia of Philosophy.

- **Tools:** Anki for spaced repetition of vocabulary, legal terms and current-affairs facts.

C Final Reminders

- (1) These procedures are scaffolding. They are not guaranteed, and they are not proven to be the best possible.
- (2) Practice and review matter more than any single technique.
- (3) Heuristics (milder, narrower, more modest) are tiebreakers. The real checks come first.
- (4) In the exam, use the passage only. In life, use everything.
- (5) Keep your error log honest. It is the most useful document you will own.